

	W	T	F	S	S
1	2	3	4	5	6
8	9	10	11	12	13
15	16	17	18	19	20
22	23	24	25	26	27

FEB 22

January 2022

Week 04

Day 018 - 347

Date 18 - 01 - 2022

information
guthrieval
DS

18

Tuesday

Trie

- ① search (word)
- ② Insert (word)
- ③ Starts with (word)

Trie

Insert

```

dried
int a[26];
bool fl;
}

```

apple
apps
apnd
bac
bat

a
b
fl=f

26 size array

apple

apps

p
fl=f

a
fl=f

c
fl=f

fl=T

fl=T

l
fl=f

fl=T

p
fl=f

Notes
e
fl=f

fl=T

Prayer should be the key of the morning and lock of the evening.

flag: True
words ends

fl=T

e
fl=f

19

Wednesday

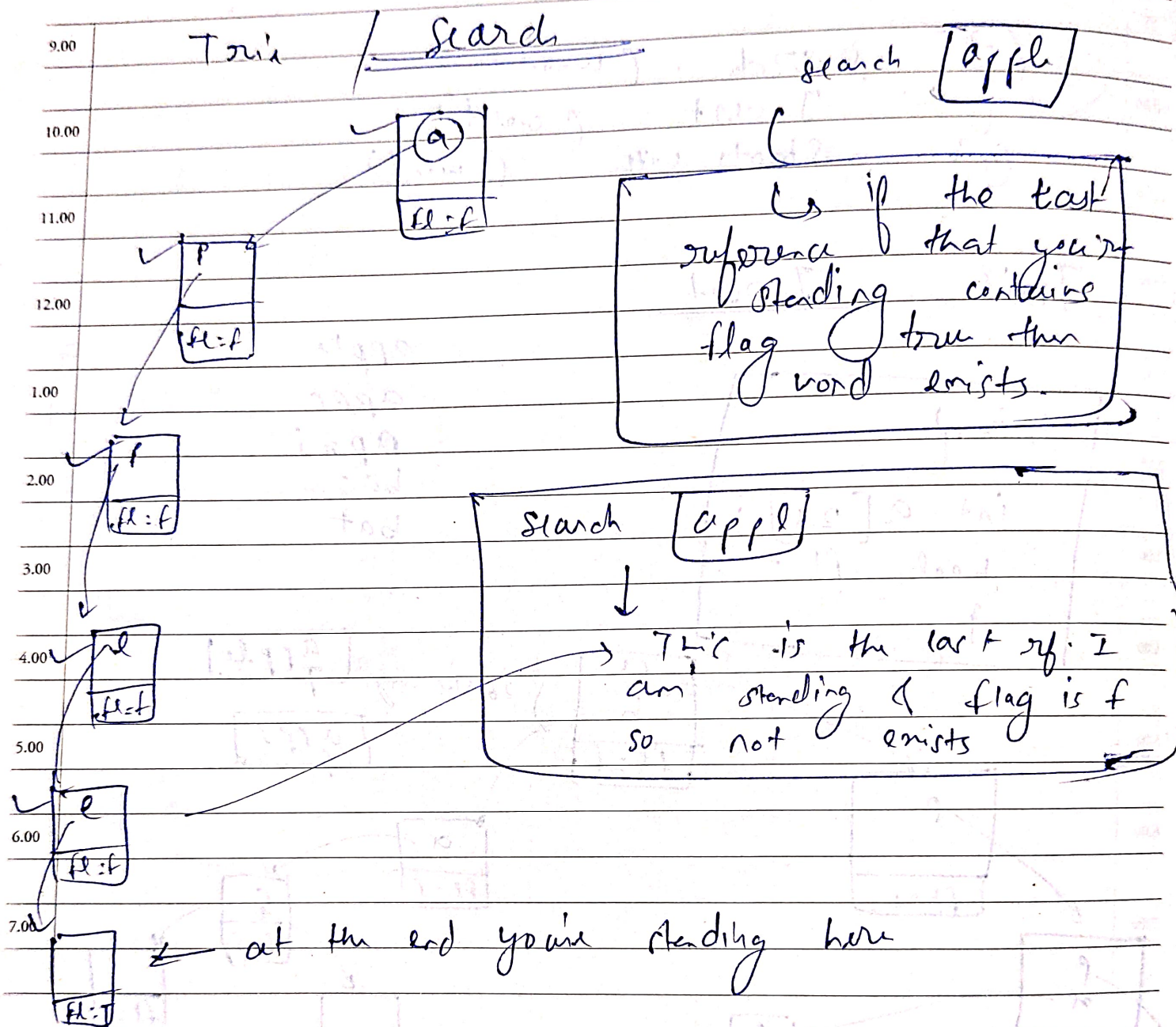
January 2022

Week 04

Day 019 - 346

Date 19 - 01 - 2022

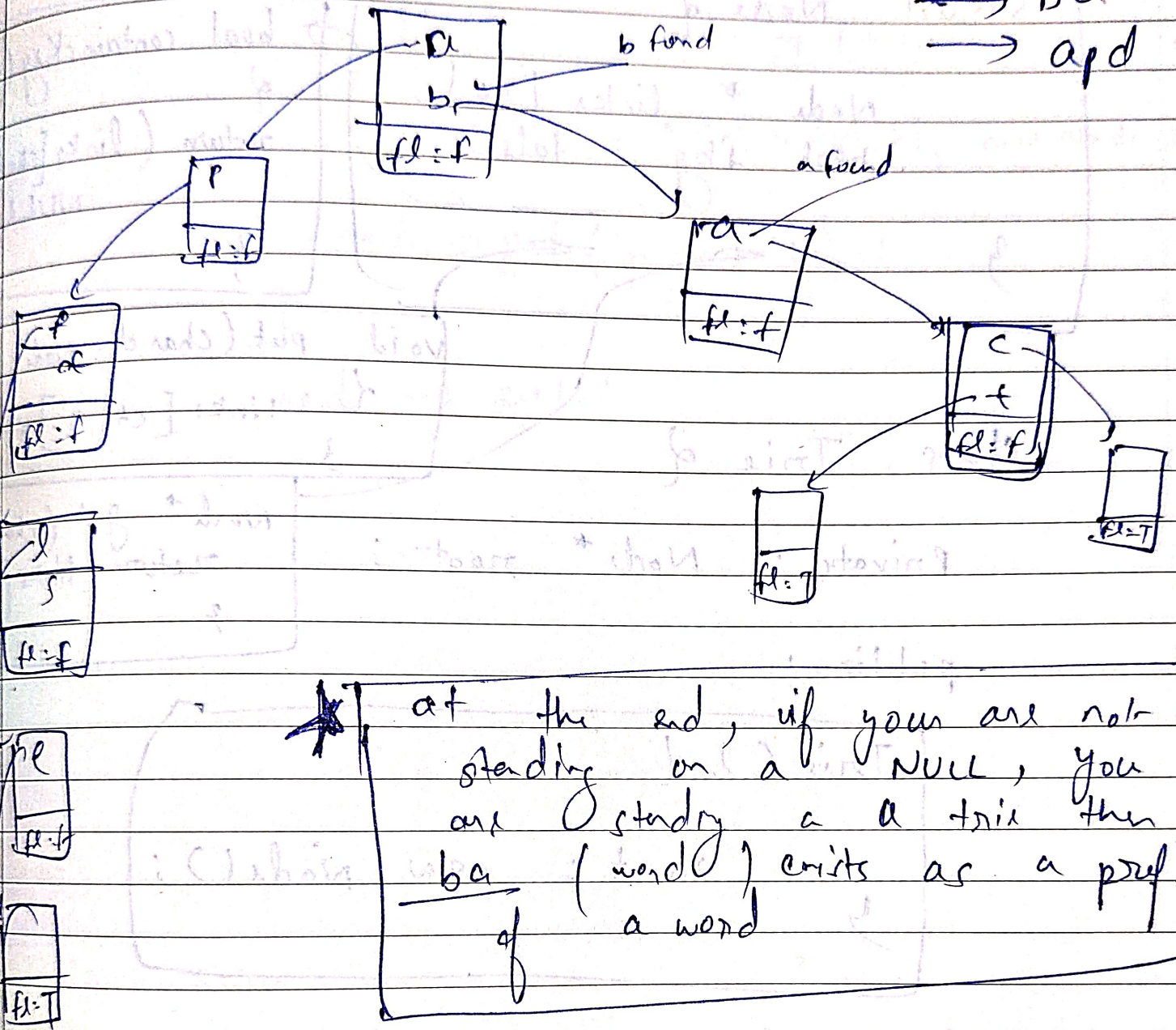
	M	T	W	T	F	S	S
JAN 22	31					1	2
	3	4	5	6	7	8	9
	10	11	12	13	14	15	16
	17	18	19	20	21	22	23
	24	25	26	27	28	29	30



Notes

The beginning is the most important part of the work.

→ bc
→ ad



at the end, if you are not standing on a null, you are standing on a trie then ba (word) exists as a prefix of a word

23

Sunday

January 2022

Week 04

Day 023 - 342

Date 23 - 01 - 2022

Let's code

	M	T	W	T	F	S	S
JAN 22	31					1	2
	3	4	5	6	7	8	9
	10	11	12	13	14	15	16
	17	18	19	20	21	22	23
	24	25	26	27	28	29	30

Struct Node

Node * links [26];

bool flag = false;

bool containsKey (char ch) {

return (links [ch - 'a'] != NULL);

}

void put (char ch, Node * node) {

links [ch - 'a'] = node;

}

Node * get (char ch) {

return links [ch - 'a'];

}

bool setEnd () {

flag = true;

}

bool isEnd () {

return flag;

}

},

Only when we are no longer afraid do we begin to live.

M	T	W	T	F	S	S
1	2	3	4	5	6	
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28						

January 2022

Week 05

Day 024 - 341

Date 24 - 01 - 2022

24

Monday

class Trie {

private : Node * root ;

public :

Trie() {

root = new Node();

}

void insert (string word) {

Node * node = root ;

for (i = 0 → word.length()) {

if (! node → containsKey (word[i])) {

node → put (word[i], new Node());

node → get (word[i]);

node → setEnd();

}

Notes

Sulking about your mistakes only leads to future ones.

25

Tuesday

January 2022

Week 05

Day 025 - 340

Date 25 - 01 - 2022

	M	T	W	T	F	S	S
JAN 22	31					1	2
	3	4	5	6	7	8	9
	10	11	12	13	14	15	16
	17	18	19	20	21	22	23
	24	25	26	27	28	29	30

```

bool search ( string word ) {
    Node * node = root ;

    for ( i=0 → word.length ( ) ) {
        if ( ! node → contains ( word [ i ] ) ) {
            return false ;
        }
        node = node → get ( word [ i ] ) ;
    }

    if ( node → isEnd ( ) == true ) {
        return true ;
    }

    return false ;
}
    
```

Notes

The first step to getting the things you want out of life is this: Decide what you want.

bool startswith (string prefix) {

Node * node = root;

for (i = 0 → word.length()) {

{ if (!node → containskey(word[i]))
 return false;
 }

node = node → get(word[i]);
 }

return true;

}

};

27

Thursday

January 2022

Week 05

Day 027 - 338

Date 27 - 01 - 2022

	M	T	W	T	F	S	S
JAN 22	31	3	4	5	6	7	1
	8	9	10	11	12	13	2
	14	15	16	17	18	19	3
	20	21	22	23	24	25	4
	26	27	28	29	30		5

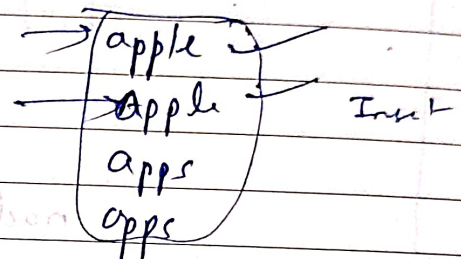
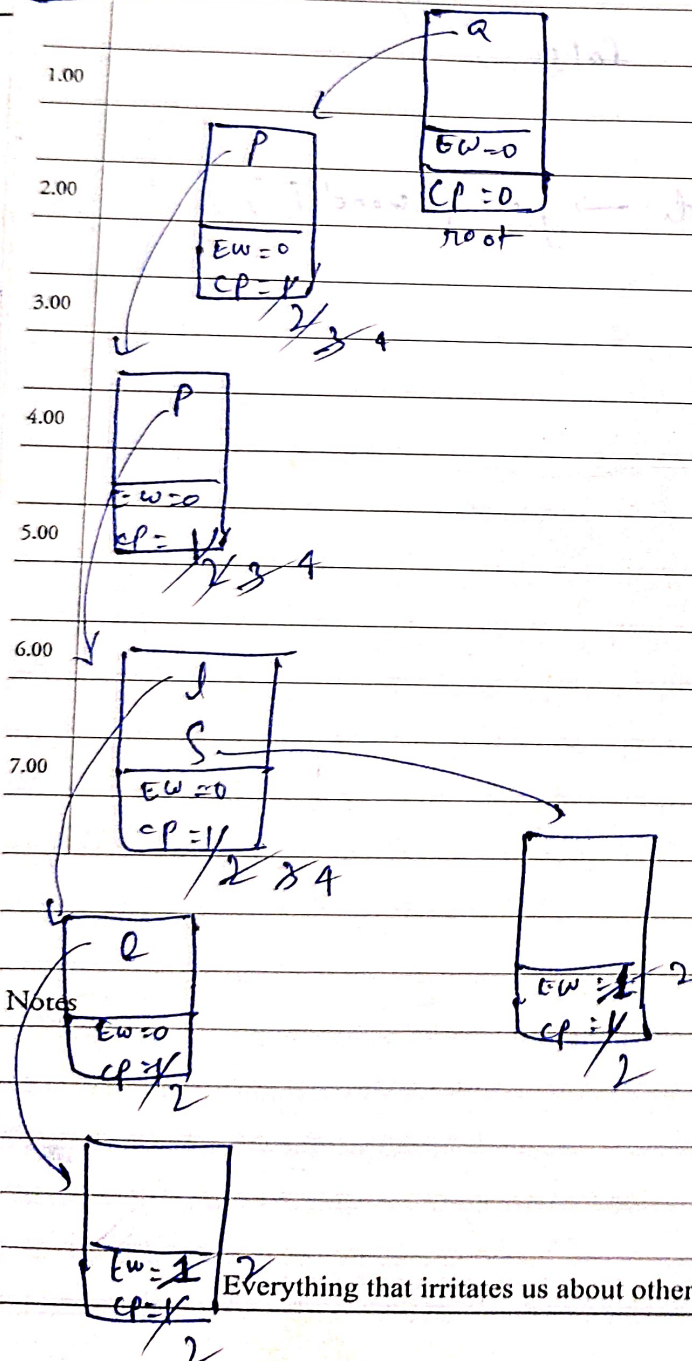
Implement Trie II

insert (word)
countWords Equal To
countWords Starts with
erase (word)

(word)
(prefix)

Implement

Input



Trie q

links [26]

ew = 0

// Ends with

cp = 0

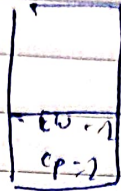
// count prefix

3

Everything that irritates us about others can lead us to an understanding of ourselves.

cont words End at

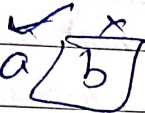
apps →



Ended at
this node

→ End, with = 2

abc →

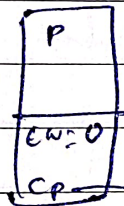


does not exist

→ no word

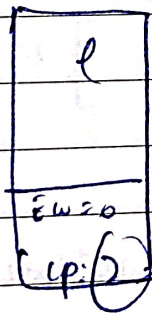
word words starts with (word)

ap → 4



I am standing at
this node, 4 cp, so 4 words
→ end, with = 4

appl → 2



2 words with prefix
appl.

29

Saturday

January 2022

Week 05

Day 029 - 336

Date 29 - 01 - 2022

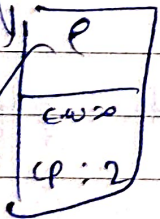
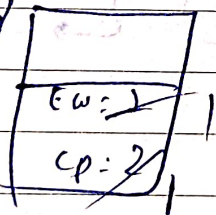
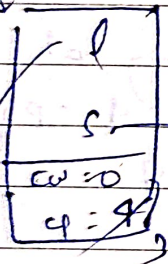
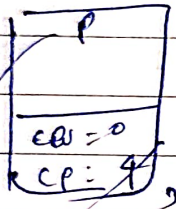
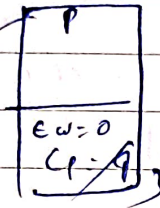
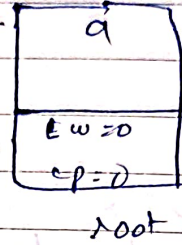
	M	T	W	T	F	S	S
JAN 22	31					1	2
	3	4	5	6	7	8	9
	10	11	12	13	14	15	16
	17	18	19	20	21	22	23
	24	25	26	27	28	29	30

Enace (around)

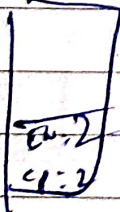
apps →

delete
"apps"

keep on
reducing
CP



Notes



When an individual or a company decides that success has been attained, progress stops.

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28						

January 2022

Week 05

Day 030 - 335

Date 30 - 01 - 2022

30

Sunday

Let's

code

Struct Node of

```
Node * links [26];
int cntEndWith = 0;
int cntPrufin = 0;
```

```
{ bool containsKey (char ch) {
    return (links[ch - 'a'] != NULL);
}
```

```
{ Node* get (char ch) {
    return links[ch - 'a'];
}
```

```
{ void put (char ch, Node* node) {
    links[ch - 'a'] = node;
}
```

```
{ void increaseEnd() {
    cntEndWith++;
}
```

```
{ void increasePrufin() {
    cntPrufin++;
}
```

Notes

Anxiety should be used as a catapult to help you move forward to objectives.

31

Monday

January 2022

Week 06

Day 031 - 334

Date 31 - 01 - 2022

	M	T	W	T	F	S	S
JAN 22	31					1	2
	3	4	5	6	7	8	9
	10	11	12	13	14	15	16
	17	18	19	20	21	22	23
	24	25	26	27	28	29	30

```
void deleteEnd () {
```

```
    cntEndwith -- ;
```

```
}
```

```
void reducePrufin () {
```

```
    cntPrufin -- ;
```

```
}
```

```
int getEnd () {
```

```
    return cntEndwith ;
```

```
}
```

```
int getPrufin () {
```

```
    return cntPrufin ;
```

```
}
```

```
};
```

Notes

The two hardest things to handle in life are failure & success.

01

Tuesday

February 2022

Week 06

Day 032 - 333

Date 01 - 02 - 2022

	M	T	W	T	F	S	S
		1	2	3	4	5	6
FEB 22	7	8	9	10	11	12	13
	14	15	16	17	18	19	20
	21	22	23	24	25	26	27
	28						

```
class Trie {
```

```
private :
```

```
Node * root ;
```

```
public :
```

```
Trie () {
```

```
root = new Node();
```

```
}
```

```
void insert (string word) {
```

```
Node * node = root ;
```

```
for (i=0 → word.length) {
```

```
if (! node → containsKey (word[i])) {
```

```
node → put (word[i], new Node());
```

```
}
```

```
node = node → get (word[i]) ;
```

```
node → insert (word[i+1]);
```

```
}
```

```
node → increaseEnd();
```

```
}
```

Notes

T	W	T	F	S	S
1	2	3	4	5	6
8	9	10	11	12	13
5	16	17	18	19	20
2	23	24	25	26	27
30	31				

February 2022

Week 06

Day 033 - 332

Date 02 - 02 - 2022

02

Wednesday

```
int countWordsEqualTo (string word) {
```

```
    Node * node = root;
```

```
    for (i=0 → word.length()) {
```

```
        if (!node → containsKey (word[i])) {
```

```
            node = node → get (word[i]);
```

```
        }
```

```
    }
```

```
    return 0;
```

```
}
```

```
node → getEnd();
```

```
}
```

The state of your life is nothing more than a reflection of your state of mind.

int countWordsStartingWith (string word) {

Node * node = root;

for (i = 0 → word.length) {

if (node → containsKey (word[i])) {

node = node → get (word[i]);

}

return 0;

}

return node → getPrefix ();

}

void erase (string word) {

Node * node = root;

for (i = 0 → word.length) {

if (node → containsKey (word[i])) {

node = node → get (word[i]);

node → reducePrefix ();

}

else return;

}

If you don't drive your business, you will be driven out of business.

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

February 2022

Week 06

Day 035 - 330

Date 04 - 02 - 2022

04

Friday

node $\rightarrow$ delete end ()

3

Trie & (lecture)

Q Given an array of numbers, & a number x .

find the max value of $(arr[i] \wedge x)$.

$\hookrightarrow$ insert all the nos (binary bits) into the trie

$\hookrightarrow$ Take x & find the max no. from array where $(no. \wedge x) \uparrow$

arr = [25, 10, 2, 8, 5, 3]

max $\rightarrow 25 \wedge 5 \Rightarrow 28 \uparrow$

$\rightarrow$ Find max val of any 2 nos with an array?

Notes

05

Saturday

February 2022

Week 06

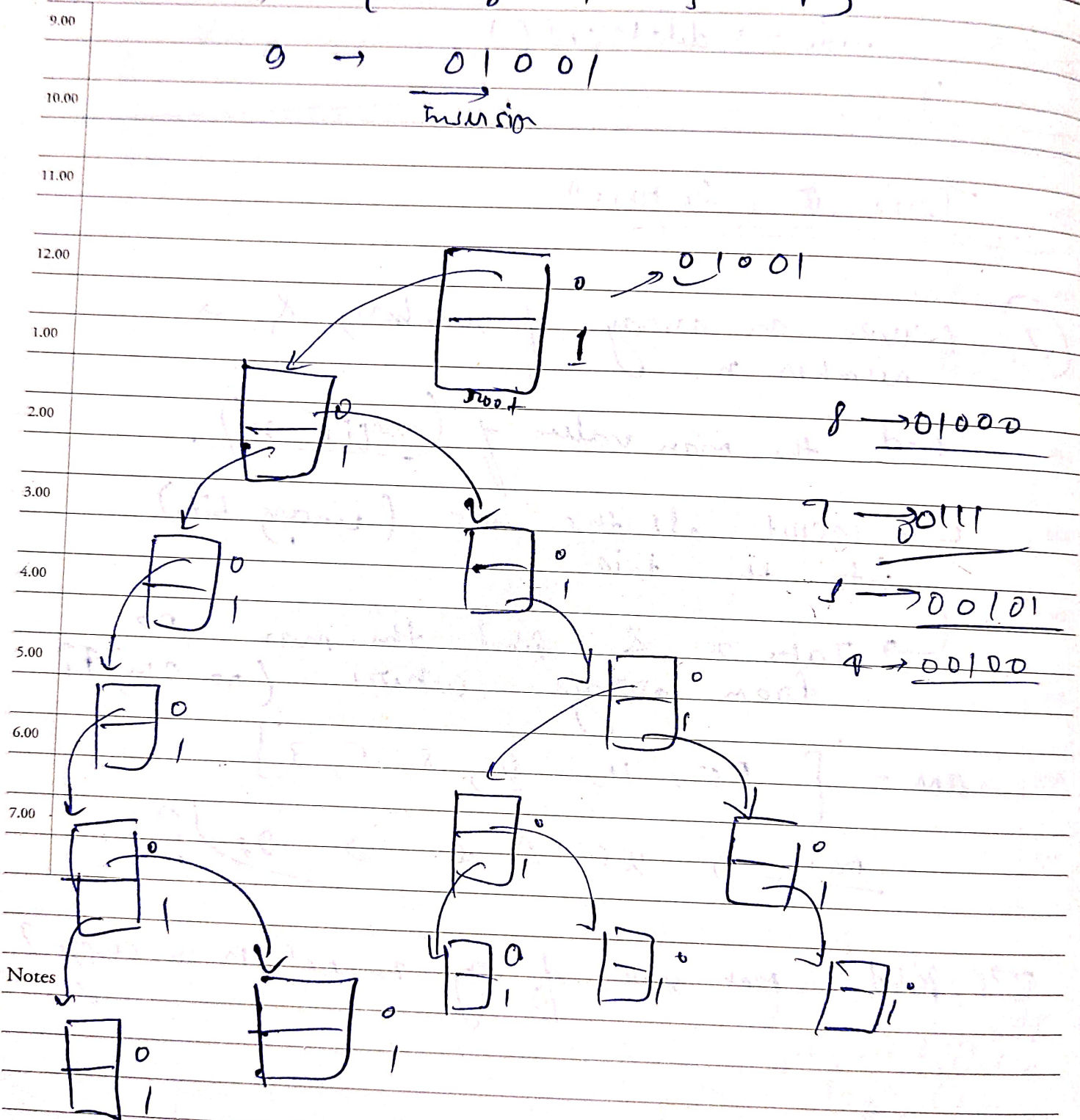
Day 036 - 329

Date 05 - 02 - 2022

	M	T	W	T	F	S	S
		1	2	3	4	5	6
FEB 22	7	8	9	10	11	12	13
	14	15	16	17	18	19	20
	21	22	23	24	25	26	27
	28						

arr → [9 8 7 5 4]

9 → 0 1 0 0 1
inversion



The person who gets the farthest is generally the one who is willing to do and dare.

M	T	W	T	F	S	S
1	2	3	4	5	6	
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

February 2022

Week 06

Day 037 - 328

Date 06 - 02 - 2022

06

Sunday

Minimizing XOR

9 8 7 5 4
 01001 01000 00111 00101 00100

if $n = 8$ 01000

initial mask

00000

→ start with the mask For last bit, if
 there a opposite if you the go
 with that bit also go with
 current bits true.

→ If opposite found → xor will be 1
 at that bit.

01000

final xor → 00111

found opposite bits 01111 → 15 15

Notes

Winning isn't everything, but wanting to win is.

07

February 2022

Week 07

Day 038 - 327

Date 07 - 02 - 2022

Monday

	M	T	W	T	F	S	S
		1	2	3	4	5	6
FEB 22	7	8	9	10	11	12	13
	14	15	16	17	18	19	20
	21	22	23	24	25	26	27
	28						

```

class Trie {
    root ;
    insert (x);
    get max (x);
}

```

```

struct Node {

```

```

    Node * links [2];

```

```

    bool containsKey (int bit) {

```

```

        return (links [bit] != NULL);
    }

```

```

    Node * get (int bit) {

```

```

        return links [bit];
    }

```

```

void put (int bit, Node * node) {

```

```

    links [bit] = node;
}

```

Keep in mind that neither success nor failure is ever final.

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

February 2022

Week 07

Day 039 - 326

Date 08 - 02 - 2022

08

Tuesday

class Trie {

private:

Node * root;

public:

Trie () {

root = new Node();

}

void insert (int num) {

for (i = 31; i >= 0; i--) {

int bit = (num >> i) & 1;

if (!node->contains (bit)) {

node->put (bit, new Node());

}

node = node->get (bit);

}

}

Notes

Give to the world the best you have and the best will come back to you.

09

Wednesday

February 2022

Week 07

Day 040 - 325

Date 09 - 02 - 2022

	M	T	W	T	F	S	S
		1	2	3	4	5	6
FEB 22	7	8	9	10	11	12	13
	14	15	16	17	18	19	20
	21	22	23	24	25	26	27
	28						

```
int getman (int num) {
```

```
    Node * node = root ;
    int mannum = 0 ;
```

```
    for (i=31 -> 0) {
```

```
        int bit = (num >> i) & 1 ;
```

```
        if (node->contains (1-bit)) {
```

```
            mannum = mannum | (1 << i) ;
```

```
            node = node -> get (1-bit) ;
```

```
        }
        else {
```

```
            node = node -> get (bit) ;
```

```
        }
    }
```

```
    return mannum ;
```

```
}
```

```
};
```

Notes

14 17
21 22 23 24
28 29 30 31

Man of Queries

Chowman

→ using offline queries

Max XOR Subarray :-

1 2 3 4 → prefix xor → Trick है दुलारी

~~दुलारी~~ Trick Max xor find कर दो
of psbl subarray ko max
xor kar